

PHYS 122: ELECTRICITY AND MAGNETISM AND VIBRATION AND WAVES

Tutorial 8

1. A light source emits visible light of two wavelengths: $\lambda = 430 \text{ nm}$ and $\lambda' = 510 \text{ nm}$. The source is used in a double-slit interference experiment in which $L = 1.50 \text{ m}$ and $d = 0.025 \text{ mm}$. Find the separation distance between the third-order bright fringes.

[Ans: $\Delta y = 0.014 \text{ m}$ or 1.44 cm]

2. In a location where the speed of sound is 354 m/s , a $2\,000\text{-Hz}$ sound wave impinges on two slits 30.0 cm apart. At what angle is the first order maximum located?

[Ans: $\theta_{\text{bright}} = 36.16^\circ$]

3. A beam of light of wavelength 550 nm travelling in air is incident on a slab of transparent material. The incident beam makes an angle of 40.0° with the normal, and the refracted beam makes an angle of 26.0° with the normal. Find the index of refraction of the material. [Ans: $n = 1.47$]

4. A narrow beam of sodium yellow light, with wavelength 589 nm in vacuum, is incident from air onto a smooth water surface at an angle of incidence of 35.0° . Determine the angle of refraction and the wavelength of the light in water.

[Ans: $\theta_2 = 25.5^\circ$ & $\lambda = 442 \text{ nm}$]

5. When an open metal pipe is cut into two pieces, the lowest resonance frequency for the air column in one piece is 256 Hz and that for the other is 440 Hz .

- a) What resonant frequency would have been produced by the original length of pipe? [Ans: $f_1 = 161.79 \text{ Hz}$]
b) How long was the original pipe? [Ans: $L = 1.06 \text{ m}$]

6. The high E string on a guitar measures 64.0 cm in length and has a fundamental frequency of 330 Hz. By pressing down so that the string is in contact with the first fret the string is shortened so that it plays an F note that has a frequency of 350 Hz. How far is the fret from the neck end of the string? [**Ans:** $L=0.037$ m]

7. Section of drainage culvert 1.23 m in length makes a howling noise when the wind blows.

- a) Determine the frequencies of the first three harmonics of the culvert if it is cylindrical in shape and open at both ends. Take $v = 343$ m/s as the speed of sound in air. [**Ans:** $f_1 = 139$ Hz, $f_2 = 278$ Hz & $f_3 = 417$ Hz]
- b) What are the three lowest natural frequencies of the culvert if it is blocked at one end? [**Ans:** $f_1 = 69.7$ Hz, $f_2 = 209.1$ Hz & $f_3 = 348.5$ Hz]
- a) For the culvert open at both ends, how many of the harmonics present fall within the normal human hearing range (20 to 20 000 Hz)? [**Ans:** $n = 143$]